LiftDrag Homework Problem

Given

- Orientation$(roll, pitch, yaw)$: $\begin{bmatrix} 0.09 & 1.48 & 0.1 \end{bmatrix} rad$
- Velocity, V_{world}: $\begin{bmatrix} 10 & 1 & 0.1 \end{bmatrix} \frac{m}{s}$
- Airfoil:
  - Profile : NACA0012 (See Appendix)
  - Area, A : $1m^2$
  - Body Basis:
    - Upward: $\begin{bmatrix} -1 & 0 & 0 \end{bmatrix}$
    - Forward: $\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$
    - Side: $\begin{bmatrix} 0 & -1 & 0 \end{bmatrix}$
  - Lateral(Side-Slip) Drag, C_{D-ss}: 0.1

Assumptions

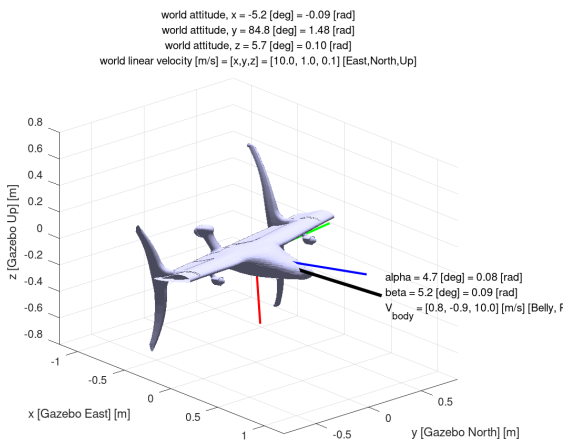
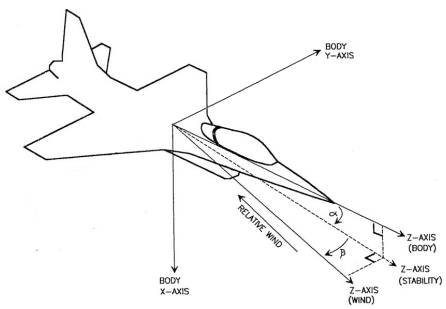
- Air Density, ρ : $1.22 \frac{kg}{m^3}$

Find

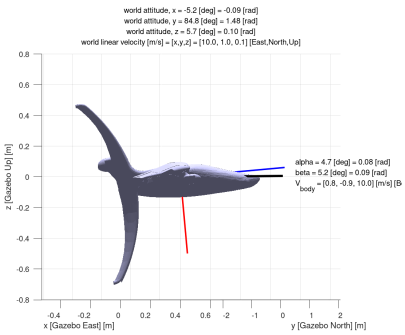
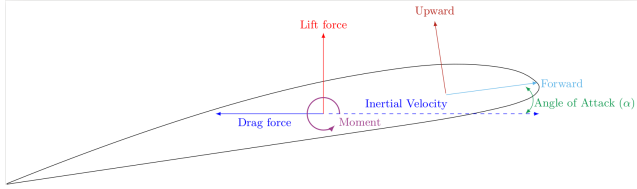
- Resultant Force On Aircraft Body, In body frame, due to:
  - Lift
  - Drag

Diagram

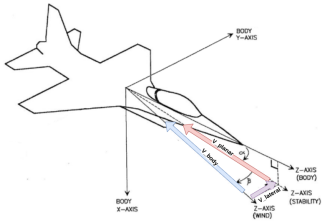
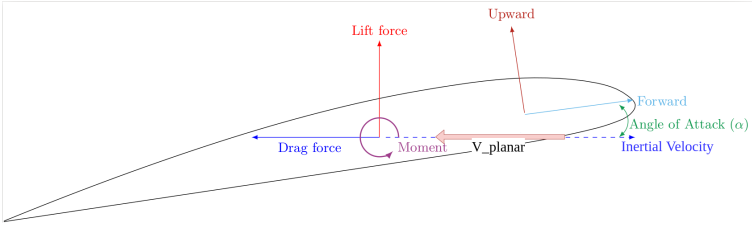
Reference

Reference	Free Body Diagram - Body Frame
	

Planar-Wind/Aerodynamic Frame

Reference	Free Body Diagram - WindFrame
	

Supplemental Vector Decomposition

Annotated Body Frame - FBD	Annotated Wind Frame - FBD
	

Analysis

Transform Velocity in World Frame to Body Frame of Aircraft

We first define the quaternion representing the rotation between the world and the aircraft,

$$Q = (q_{yaw} q_{pitch} q_{roll})$$

See Appendix B , for composing a quaternion from euler angles, but using any modern calculator will resolve to the following:

$$Q_{world \to body} = [\,0.7353 \quad -0.06681i \quad 0.6711j \quad 0.06719k\,]$$

$$Q^{-1}_{world \to body} = [\,0.7353 \quad 0.06681i \quad -0.6711j \quad -0.06719k\,]$$

Using the quaternion rotation and its inverse, we can now calculate V_{body} using the following Hamiltonian rotations.

$$V_{body} = Q_{world \to body}(V_{world} * Q^{-1}_{world \to body})$$

Again, most devices now support quaternion-vector multiplication, using such functions, this resolves the V_{body} to be:

$$V_{body} = [\,0.81165 \quad -0.90368 \quad 9.97673\,]\, m/s$$

Transform Velocity In Body Frame to Planar-Wind/Aerodynamic Frame

Referencing the aerodynamic frame diagrams above, the velocity in body is broken down into two planar wind frames:

- The plane parallel to the airfoil profile, V_{planar}
- The plane, $V_{lateral}$

Reorient vector in body to wind:

$$V_{wind} = \{V_{body} \cdot Upward, V_{body} \cdot Side, V_{body} \cdot Forward\}$$
$$V_{wind} = \{[0.81165 \quad -0.9037 \quad 9.9767] \cdot [-1 \quad 0 \quad 0], [0.81165 \quad -0.9037 \quad 9.9767] \cdot [0 \quad 0 \quad 1], [0.81165 \quad -0.9037 \quad 9.9767] \cdot [0 \quad -1 \quad 0]\}$$
$$V_{wind} = [-0.81165 \quad 0.9037 \quad 9.9767]$$

Magnitude of Velocity Vector, V_∞

$$V_\infty = ||V_{body}|| = \sqrt{{V_{body}}_x{}^2 + {V_{body}}_y{}^2 + {V_{body}}_z{}^2}$$
$$V_\infty = \sqrt{0.8117^2 + -0.9037^2 + 9.9767^2}$$
$$V_\infty = 10.0504$$

Planar Velocity

$$V_{planar} = \sqrt{{V_{wind}}_x{}^2 + {V_{wind}}_z{}^2}$$

$$V_{planar} = \sqrt{(0.81165\frac{m}{s})^2 + (9.97673\frac{m}{s})^2}$$

$$V_{planar} = 10.01\frac{m}{s}$$

Determine angle of attack, α

$$\alpha = tan^{-1}(\frac{-V_{wind\,x}}{V_{wind\,z}})$$

$$\alpha = tan^{-1}(\frac{-0.8117}{9.9767})$$

$$\alpha = 4.651^\circ$$

Lateral Velocity

$$V_{lateral} = -V_{wind\,y}$$
$$V_{lateral} = -0.9037$$

$$\beta = sin^{-1}(\frac{V_{lateral}}{V_\infty})$$
$$\beta = sin^{-1}(\frac{(-0.9037)}{10.0504})$$
$$\beta = -5.1587^\circ$$

Calculate Lift And Drag due to Planar Velocity

Planar dynamic pressure then is defined as

$$q_{planar} = \frac{1}{2}\rho V_{planar}^2$$

$$q_{planar} = \frac{1}{2}(1.22)(10.01)^2[\frac{kg}{m^3}][\frac{m^2}{s^2}]$$

$$q_{planar} = 61.122\frac{kg}{m \cdot s^2}$$

Note: $\frac{kg}{m \cdot s^2} = 1Pa$, confirming pressure through unit analysis

Calculating Lift in Aerodynamic Frame:

Using α and linear interpolation with Appendix A:

$$CL = 0.511614$$

$$Lift = C_L q_{planar} A$$
$$Lift = 0.5116 \cdot 61.122 \cdot 1[\frac{kg}{m \cdot s^2}][m^2]$$
$$Lift = 31.269N$$

Calculating Drag in Aerodynamic Frame:

Using α and linear interpolation with Appendix A:

$$CD = 0.013442$$

$$Drag = C_D q_{planar} A$$
$$Drag = 0.0134 \cdot 61.122 \cdot 1[\frac{kg}{m \cdot s^2}][m^2]$$
$$Drag = 0.821N$$

Calculate Drag due to Lateral Velocity

For the purposes of calculate drag due to side slip, we use sideslip dynamic pressure.

$$q_{ss} = \frac{1}{2}\rho {V_{body_y}}^2$$
$$q_{ss} = \frac{1}{2}(1.22)(-0.90368)^2[\frac{kg}{m^3}][\frac{m^2}{s^2}]$$
$$q_{ss} = 0.498\frac{kg}{m \cdot s^2}$$

$$Drag_{ss} = C_{D-ss} q_{ss} A_{lateral}$$
$$Drag_{ss} = 0.1 \cdot 0.498 \cdot 0.1[\frac{kg}{m \cdot s^2}][m^2]$$
$$Drag_{ss} = 0.005N$$

Transform From Aerodynamic Frame Back to Body Frame

Rotating vectors from wind frame back into body frame we use basic trigonometry (refer to figure 2)

$$F_{up} = L * cos(\alpha) + D * sin(\alpha)$$
$$F_{fwd} = L * sin(\alpha) - D * cos(\alpha)$$
$$F_{up} = 31.269 * cos(4.651^\circ) + 0.821N * sin(4.651^\circ)$$
$$F_{fwd} = 31.269 * sin(4.651^\circ) - 0.821N * cos(4.651^\circ)$$
$$F_{up} = 31.232N$$
$$F_{fwd} = 1.716N$$

$$F_{lat} = sign(\beta) * Drag_{ss}$$
$$F_{lat} = sign(-5.1587) * 0.005$$
$$F_{lat} = -0.005$$

For the calculation of the final resultant force in body-fixed coordinate frame, convert the scalar force quantities into the appropriate directions in body-fixed coordinate frame.

$$F_R = (F_{up} * \vec{u}_{up}) + (F_{fwd} * \vec{u}_{fwd}) + (F_{lat} * \vec{u}_{lat})$$
$$F_R = (31.232 * [\,-1 \quad 0 \quad 0\,]) + (1.716 * [\,0 \quad 0 \quad 1\,]) + (-0.005 * [\,0 \quad -1 \quad 0\,])$$
$$F_R = [\,-31.2324 \quad 0.005 \quad 1.716\,]\,N$$

Result

$$ResultantForceonBody : [\,-31.2324 \quad 0.005 \quad 1.716\,]\,N$$

Appendix A: NACA0012 Lookup Table

Angle of Attack$[^\circ]$	C_L	C_D
-180.0000	-0.0000	0.02500
-175.00000	0.69000	0.05500
-170.00000	0.85000	0.14000
-165.00000	0.67500	0.23000
-160.00000	0.66000	0.32000
-155.00000	0.74000	0.42000
-150.00000	0.85000	0.57000
-145.00000	0.91000	0.75500
-140.00000	0.94500	0.92500
-135.00000	0.94500	1.08500
-130.00000	0.91000	1.22500
-125.00000	0.84000	1.35000
-120.00000	0.73500	1.46500
-115.00000	0.62500	1.55500
-110.00000	0.51000	1.63500
-105.00000	0.37000	1.70000
-100.00000	0.22000	1.75000
-95.00000	0.07000	1.78000
-90.00000	-0.07000	1.80000
-85.00000	-0.22000	1.80000
-80.00000	-0.37000	1.78000
-75.00000	-0.51500	1.73500
-70.00000	-0.65000	1.66500
-65.00000	-0.76500	1.57500
-60.00000	-0.87500	1.47000
-55.00000	-0.96500	1.34500
-50.00000	-1.04000	1.21500
-45.00000	-1.08500	1.07500
-40.00000	-1.07500	0.92000
-35.00000	-1.02000	0.74500

Angle of Attack$[^\circ]$	C_L	C_D
-30.00000	-0.91500	0.57000
-27.00000	-0.96460	0.47300
-26.00000	-0.91090	0.44600
-25.00000	-0.85720	0.42000
-24.00000	-0.80340	0.39400
-23.00000	-0.74970	0.36900
-22.00000	-0.69560	0.34400
-21.00000	-0.64140	0.32000
-20.00000	-0.58700	0.29700
-19.00000	-0.53220	0.27400
-18.00000	-0.47680	0.25200
-17.00000	-0.42000	0.23100
-16.00000	-0.36200	0.21000
-15.00000	-0.30820	0.19000
-14.00000	-0.25460	0.17100
-13.00000	-0.20300	0.15200
-12.00000	-0.15330	0.13400
-11.00000	-0.10950	0.07600
-10.00000	-0.13250	0.01880
-9.00000	-0.85270	0.02030
-8.00000	-0.82740	0.01850
-7.00000	-0.74600	0.01700
-6.00000	-0.66000	0.01520
-5.00000	-0.55000	0.01400
-4.00000	-0.44000	0.01240
-3.00000	-0.33000	0.01140
-2.00000	-0.22000	0.01080
-1.00000	-0.11000	0.01040
0.00000	0.00000	0.01030
1.00000	0.11000	0.01040

Angle of Attack$[^\circ]$	C_L	C_D
2.00000	0.22000	0.01080
3.00000	0.33000	0.01140
4.00000	0.44000	0.01240
5.00000	0.55000	0.01400
6.00000	0.66000	0.01520
7.00000	0.74600	0.01700
8.00000	0.82740	0.01850
9.00000	0.85270	0.02030
10.00000	0.13250	0.01880
11.00000	0.10950	0.07600
12.00000	0.15330	0.13400
13.00000	0.20300	0.15200
14.00000	0.25460	0.17100
15.00000	0.30820	0.19000
16.00000	0.36200	0.21000
17.00000	0.42000	0.23100
18.00000	0.47680	0.25200
19.00000	0.53220	0.27400
20.00000	0.58700	0.29700
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22.00000	0.69560	0.34400
23.00000	0.74970	0.36900
24.00000	0.80340	0.39400
25.00000	0.85720	0.42000
26.00000	0.91090	0.44600
27.00000	0.96460	0.47300
30.00000	0.91500	0.57000
35.00000	1.02000	0.74500
40.00000	1.07500	0.92000
45.00000	1.08500	1.07500

Angle of Attack$[^\circ]$	C_L	C_D
50.00000	1.04000	1.21500
55.00000	0.96500	1.34500
60.00000	0.87500	1.47000
65.00000	0.76500	1.57500
70.00000	0.65000	1.66500
75.00000	0.51500	1.73500
80.00000	0.37000	1.78000
85.00000	0.22000	1.80000
90.00000	0.07000	1.80000
95.00000	-0.07000	1.78000
100.00000	-0.22000	1.75000
105.00000	-0.37000	1.70000
110.00000	-0.51000	1.63500
115.00000	-0.62500	1.55500
120.00000	-0.73500	1.46500
125.00000	-0.84000	1.35000
130.00000	-0.91000	1.22500
135.00000	-0.94500	1.08500
140.00000	-0.94500	0.92500
145.00000	-0.91000	0.75500
150.00000	-0.85000	0.57000
155.00000	-0.74000	0.42000
160.00000	-0.66000	0.32000
165.00000	-0.67500	0.23000
170.00000	-0.85000	0.14000
175.00000	-0.69000	0.05500
180.00000	0.00000	0.02500

Appendix B: Euler Angles to Quaternion

Translating Euler angles into a quaternion representation of the rotation follows:

Assuming the angles for roll, pitch, and yaw are provided in radians, then it follows that

Given:

$$\phi = \frac{roll}{2}$$
$$\theta = \frac{pitch}{2}$$
$$\psi = \frac{yaw}{2}$$

Then

$$Q_w = cos(\phi) * cos(\theta) * cos(\psi) + sin(\phi) * sin(\theta) * sin(\psi)$$
$$Q_x = sin(\phi) * cos(\theta) * cos(\psi) - cos(\phi) * sin(\theta) * sin(\psi)$$
$$Q_y = cos(\phi) * sin(\theta) * cos(\psi) + sin(\phi) * cos(\theta) * sin(\psi)$$
$$Q_z = cos(\phi) * cos(\theta) * sin(\psi) - sin(\phi) * sin(\theta) * cos(\psi)$$

Appendix C: Hamiltonian Vector-Quaternion Multiplication

Given some vector V_a and a reference given by the quaternion $Q = Q_{a \to b}$, the Vector represented in the reference frame is defined as:

$$V_b = Q \cdot V_a \cdot Q^{-1}$$

This is solved by translating V_a into a quaternion Q_V defined as:

$$Q_V = 0 + V_x i + V_y j + V_z k$$

From there the hamiltonian products reduce down into the following condensed form:

$$\textit{Given a Vector U, such that:}$$
$$U = [\, Q_x \quad Q_y \quad Q_z \,]$$
$$V_b = 2(U \cdot V_a)U + (Q_w^2 - U \cdot U)V_a + 2Q_w(U \times V_a);$$

Alternatively, this reduces further down to a more discrete form as follows:

$$V_{b-x} = 2(Q_w V_{a-z} Q_y + Q_x V_{a-z} Q_z - Q_w V_{a-y} Q_z + Q_x V_{a-y} Q_y) + V_{a-x}(Q_w^2 + Q_x^2 - Q_y^2 - Q_z^2)$$
$$V_{b-y} = 2(Q_w V_{a-x} Q_z + Q_x V_{a-x} Q_y - Q_w V_{a-z} Q_x + Q_y V_{a-z} Q_z) + V_{a-y}(Q_w^2 - Q_x^2 + Q_y^2 - Q_z^2)$$
$$V_{b-z} = 2(Q_w V_{a-y} Q_x - Q_w V_{a-x} Q_y + Q_x V_{a-x} Q_z + Q_y V_{a-y} Q_z) + V_{a-z}(Q_w^2 - Q_x^2 - Q_y^2 + Q_z^2)$$